Polymarket Global: Two-Sided Program Plan & Audit Report

Branch: pmus-rebuild | Commit: b4b8f9c | Venue: Polymarket Global CLOB | Date: October 11, 2026

Status: Fully Implemented & Hardened (195 Unit Tests Passing)	**Target Venue:** Polymarket Global (clob.polymarket.com)
Live VPS Status: Single-Sided Active (Unmodified, 0 Downtime)	**Key Feature:** Two-Sided vs Single-Sided Automatic Optimizer

1. Mathematical Analysis: Two-Sided vs Single-Sided Yield

Polymarket Global liquidity rewards calculate maker score as S = ((v - s) / v)^2 * size. With midpoint in [0.10, 0.90], single-sided quotes receive a 1/3 score divisor, while two-sided quotes receive full 1.0 scoring (3x multiplier per contract). However, capital efficiency depends on dollar cost, not contract count:

Parameter	Single-Sided (Netanyahu YES @ $0.39)	Two-Sided Pair (YES @ $0.39 + NO @ $0.60)
Capital Requirement	$0.39 per contract	~$0.99 per contract pair
Affordable Contracts ($80 Budget)	**205 contracts** ($79.95 deployed)	**80 contract pairs** ($79.20 deployed)
Scoring Formula	S_bid / 3 (1/3 weight)	min(S_yes, S_no) + |S_yes - S_no|/3 (full weight)
Estimated Daily Reward	**~$0.80 - $1.20 / day** (Higher dollar return)	**~$0.35 - $0.45 / day** (Lower dollar return)
Directional Risk	Unhedged: holds 205 YES contracts if filled	**Hedged / Delta-Neutral:** holds YES + NO = $1.00 settlement

Strategic Conclusion: At $85 budget, single-sided quoting buys ~2.5x more contracts and generates higher dollar yields. Two-sided quoting's primary advantage at this capital scale is **risk mitigation (delta-neutral hedging)** rather than higher raw earnings.

2. Audit & Hardening Fixes Applied

Four critical architectural hardening fixes were implemented and verified with unit tests:

Component	Vulnerability Addressed	Hardened Behavior
Paired Quoter Partial Fills (lip_quoter.py)	When net > 0, re-posting full quote_size caused over-buying and risk gate lockouts.	Only quotes min(target_sz - net, max_contracts) if remaining >= min_size. Otherwise cleanly stops bidding.
Deterministic Pair Tracking (condition_selector.py)	Pair scoring previously relied on dictionary insertion order in engine._pairs.	AllocationResult explicitly records condition_pairs tuples. Grouping helper removed.
Buying Power Gating (engine.py)	Reallocation every 30m re-evaluated against total budget even if capital was locked in inventory.	LiveEngine.prepare queries live buying power from exchange and caps allocation budget accordingly.
Configurable Preference (config.py & selector)	Two-sided was previously ignored whenever single-sided offered $0.01 higher expected reward.	PMUS_PAIR_PREFERENCE setting allows weighting two-sided hedged pairs when desired.

3. Verification Results & Deployment Options

Test Suite: 195 passing unit tests (0 failures). | **Live VPS:** Quoting Netanyahu YES @ 0.39 with 0 downtime.

Option	Configuration	Expected Behavior & Rationale
Option A (Recommended) Maintain Current Run	Leave live VPS running as-is until Oct 12 payout verification.	Inspect client.get_total_earnings_for_user_for_day('2026-10-11') tomorrow to confirm ground-truth payout before modifying live production.
Option B Deploy Optimized Selector	Deploy b4b8f9c with PMUS_PAIR_PREFERENCE=1.0	Selects the mathematically optimal yield per dollar across all markets (will quote Netanyahu single-sided on $85; auto-switches to pairs on larger capital).
Option C Deploy Forced Hedged Pairs	Deploy b4b8f9c with PMUS_PAIR_PREFERENCE=0.5	Forces selection of delta-neutral two-sided pairs (e.g. Bernie Sanders or OpenSea), sacrificing ~50% dollar yield for zero directional price risk.